

Luke Pitstick

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EDUCATION

University of Colorado Boulder

Bachelor of Arts in Computer Science and Political Science

Aug. 2024 – May 2028

Boulder, CO

WORK EXPERIENCE

AI/ML Engineer Intern

WattByte Nexus

Jun. 2026 – Present

Golden, CO

- Design and implement physics-informed neural networks with PyTorch to predict and reduce wildfire risk from power-grid infrastructure.
- Build event-driven wildfire-risk simulations using NOAA weather data, AWS, and Redis to model grid assets, environmental conditions, and failure scenarios.
- Design and build an AWS pipeline using Fargate, SQS, and Glue to process video footage into segmented 3D representations.

Junior Data Manager

National Oceanic and Atmospheric Administration

Oct. 2025 – Present

Boulder, CO

- Automate ingestion, validation, and archival workflows for 10TB+/week of ocean acoustic data using Python, Oracle DB, and AWS S3.
- Maintain end-to-end sonar data workflows ingesting water column and passive acoustic data from NOAA research vessels into NCEI's national archive.
- Develop Python tools to automate QC and enforce metadata standards across oceanographic data submissions to NOAA/NCEI.

Data Science Intern

Ranked Choice Voting for Longmont

Aug. 2025 – Dec. 2025

Boulder, CO

- Engineered high-resolution demographic maps in ArcGIS, R, and Python, enabling campaign teams to identify turnout gaps and execute targeted outreach.
- Developed fixed-effects regression models to predict voter turnout, allowing the campaign team to make targeted decisions.

LEADERSHIP EXPERIENCE

Representative-at-Large & Press Secretary

University of Colorado Student Government

May 2025 – May 2026

Boulder, CO

- Supported allocation of a \$36M student budget across 12 cost centers, coordinating with university administrators to ensure equitable fund distribution.
- Led a 6-person team to increase student engagement across social media channels by 15%.

PROJECTS

Recipe Social Media App | [React Native](#), [Python](#), [pgvector](#), [Figma](#) | [Launch Page](#)

June 2026 – Present

- Develop and design a social media recipe app that allows users to discover new recipes and plan their week.
- Build a personalized recipe recommendation engine using vector embeddings and pgvector to learn user preferences from swipe behavior.
- Develop a data pipeline in order to web scrape recipes, clean, and convert them to vector embeddings in order to deliver a wide variety of new content.

InfraDrone | [Python](#), [C++](#), [PyTorch](#), [TensorRT](#) | [GitHub](#)

Apr. 2026 - Present

- Design and build an autonomous drone and ground station using a Jetson Orin Nano, MAVLink, and TensorRT for offline road-damage surveys.
- Developed a custom AI pipeline to detect, classify, and calculate the severity of road damage using augmented YOLOv26 models resulting in 80% precision and 75% mAP@50.
- Implement a temporal engine to determine cracks growth over time using YOLO segmentation models and a custom GPS-based algorithm.

Renewably | [Python](#), [XGBoost](#), [Polars](#), [React](#), [FastAPI](#) | [Demo](#)

Apr. 2026

- Won 1st place at BlasterHacks 2026; built an interactive renewable energy siting platform with React and ArcGIS.
- Trained an XGBoost model on geospatial features to score wind-farm site viability against existing wind turbine locations. Achieved 96% accuracy on validation data.
- Built a FastAPI optimization pipeline to evaluate thousands of candidate locations within user-defined regions.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, R

Frameworks/Libraries: PyTorch, NumPy, scikit-Learn, pandas, Polars, ggplot2, Matplotlib, React, FastAPI

Cloud/Databases: Redis, AWS, GCP, PostgreSQL, Modal, Render

Other: Figma, ArcGIS, Bash, Linux, Git